

Chaitanya Vemula
Senior Data Engineer

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Professional Summary:

- ✓ Data Engineer with 05+ years of IT experience, Designed and implemented Cloud ETL solutions using **Azure Data Factory, Azure Data Lake Storage Gen2, Azure Databricks** and **Azure Synapse Analytics** and Data Warehousing Solutions.
- ✓ Hands - on experience in **Azure Cloud Services, Azure SQL Server, Azure Logic Apps, Azure Functions, Azure Monitor, Key Vault, Azure Event hubs, Azure Cosmos DB, Azure Stream Analytics.**
- ✓ Developed **Azure Data Factory (ADF)** batch pipelines to ingest and transform data from relational sources into **Azure Data Lake Storage (ADLS Gen2)**, implementing incremental data loads and ETL processes to clean and load data into **Delta tables**.
- ✓ Expert in leveraging **Azure Data Factory** to design and implement scalable ETL pipelines that automate data ingestion and transformation, resulting in improved data accessibility and enhanced decision-making processes.
- ✓ Proficient in utilizing **Azure Databricks** for collaborative data processing, employing Apache Spark to optimize large-scale data transformations and analytics, leading to significant performance improvements and actionable insights.
- ✓ Skilled in architecting and managing data storage solutions using **Azure Data Lake Storage Gen2**, ensuring efficient storage of unstructured and structured data while maintaining security and compliance across data assets.
- ✓ Experienced in developing end-to-end analytics solutions using **Azure Synapse Analytics**, integrating data from multiple sources and enabling advanced analytics capabilities to drive business intelligence and data-driven decision-making.
- ✓ Involved in development of roadmaps and deliverables to advance the migration of existing solutions on-premises systems/applications to **Azure Cloud**.
- ✓ Developed Data pipelines, **Dataflows, Event Streams**, Lakehouse, ML Models, Notebooks, Reports, Spark jobs and warehouse in Synapse Data engineering on **Microsoft Fabric**.
- ✓ Experience in Developing Spark applications using **Spark - SQL** in **Databricks** for data extraction, transformation and aggregation from multiple file formats for analysing & transforming the data to uncover insights into the customer usage patterns.
- ✓ Developed real-time streaming solutions using **Apache Kafka** and **Azure Event Hubs**, integrated with **Azure Databricks** for stream processing using **PySpark**.
- ✓ Extensive knowledge and experience in dealing with **Relational Database Management** Systems Normalization, Stored Procedures, Constraints, Joins, Indexes, Data Import/Export, Triggers.
- ✓ Expert at data transformations like lookups, Derived Column, Conditional Splits, Sort, Data Conversation, Multicast and Derived columns, Union All, Merge Joins, Merge, Fuzzy Lookup, Fuzzy Grouping, Pivot, Un - pivot and SCD to load data in SQL SERVER destination.
- ✓ Developed and optimized data processing applications across **Teradata, Oracle, SQL Server, and MySQL databases**. streamlined query execution time by 40% through efficient indexing strategies in high-volume environments.
- ✓ Experience in Extraction, Transformation and Loading (ETL) data from various sources into Data Warehouses, as well as data processing like collecting, aggregating, and moving data from various sources using **Apache Flume, Kafka**.
- ✓ Developed iterative algorithms using **Spark Streaming** in Scala and Python to builds near real-time dashboards.
- ✓ Experienced in loading data to Hive partitions and created buckets in **Hive** and developed **Map Reduce** jobs to automate transfer the data from **H-Base**.
- ✓ Extensive experience in processing and transforming diverse file formats, including **CSV, JSON, XML, ORC, Avro, and Parquet**, utilizing **Spark** with **Python, PySpark, and Scala** to enhance data analysis and development workflows.
- ✓ Strong expertise across the technology stack, encompassing ETL, data analysis, cleansing, matching, quality, audit, and design, while also possessing extensive experience in designing and optimizing **OLAP** and **OLTP** systems for high-performance database environments.

- ✓ Created interactive **dashboards and reports in Power BI**, leveraging data from various sources for insightful business intelligence and reporting.
- ✓ Scheduled and monitored data workflows with **Control-M** and **Apache Airflow** for coordinated execution of complex tasks.
- ✓ Leveraged **SharePoint** and **Confluence** for centralized documentation and project collaboration, enhancing team communication and improving accessibility to project plans and technical specifications.
- ✓ Developed troubleshooting guides for Azure Data Factory, Databricks, and SQL-based ETL processes, reducing incident resolution time by 30%.
- ✓ Ensured adherence to testing best practices, including unit, integration, and end-to-end testing, to validate data accuracy, reliability, and performance in ETL workflows, improving data integrity across systems.
- ✓ Involved in all the phases of **Software Development Life Cycle** (Requirements Analysis, Design, Development, Testing, Deployment, and Support) and **Agile methodologies**.
- ✓ Leveraged **Jira** for **Agile project management**, facilitating **Scrum** ceremonies such as daily stand-ups, sprint planning, and retrospectives, which enhanced team collaboration and transparency; conducted retrospectives to identify actionable insights for continuous improvement.

Technical Skills:

Azure Services:	Azure Data Factory, Azure Data Bricks, Logic Apps, Functional App, Azure DevOps, Azure Synapse Analytics, Key Vault, Azure Event Hub, Azure Monitor, Azure Blob Storage, Azure Functions, MS Fabric.
Databases:	Snowflake, MS SQL Server 2016/2014/2012, Azure Data Lake Storage Gen2, Azure SQL DB, MS Excel, MS Access, Oracle 11g/12c, Redshift, Azure Synapse, Cosmos DB, Mongo DB PostgreSQL, MySQL
Big Data Technologies:	MapReduce, Hive, Teg, Python, PySpark, Scala, Kafka, Spark streaming, Oozie, Sqoop, HDFS, Flume, Zookeeper
Hadoop Distribution:	Cloudera, Horton Works, Apache
Languages:	Python, SQL, PL/SQL, Java, HiveQL, Scala, Shell Scripting, Bash, PowerShell
Web Technologies:	HTML, CSS, JavaScript, XML, JSP, Restful, SOAP
Operating Systems:	Windows (XP/7/8/10), UNIX, LINUX, UBUNTU, CENTOS
Build Automation tools:	Ant, Maven, Jenkins
Version Control:	GIT, GitHub, Bitbucket
Methodology:	Agile, Scrum
IDE & Build Tools, Design:	Eclipse, Visual Studio, PyCharm, IntelliJ IDEA, VS Code

Education:

- ✓ Masters in Computer science at University of Bridgeport, Connecticut, USA in 2024
- ✓ Completed Bachelors in Computer science from ANU, Guntur in 2017

Certifications:

- ✓ Azure Data Engineer Associate (DP-203)
- ✓ Azure Fundamentals (AZ-900)

Professional Experience:

Client: Casey's, Des Moines, IA

Feb 2025 to Present

Role: Senior Data Engineer

Responsibilities:

- ✓ Designed and implemented end-to-end ETL pipelines using **Azure Data Factory**, incorporating Copy Activities, Lookup activity, Mapping Data Flows, For Each, Set Variable, Control Flow Activities, Get Meta data Activity and Web activity which improved in data transferring efficiency.
- ✓ Managed data orchestration with **Azure Data Factory**, scheduling workflows to integrate Databricks and Synapse Analytics activities, ensuring seamless data processing across platforms.
- ✓ Optimized data transformation workflows with **Azure Databricks**, leveraging PySpark DataFrames, RDDs, and Databricks Runtime to enhance data processing performance by 60%.

- ✓ Developed robust ETL pipelines using **Azure Synapse Analytics** and **Databricks** for efficient data ingestion, transformation, and loading from various data sources into **Azure Data Lake Storage** and **SQL databases**.
- ✓ Created robust data pipelines in **Azure Synapse Analytics**, using Data Flows for code-free ETL and Databricks **Notebooks** for complex transformations.
- ✓ Built and managed **Delta Lake tables** in Databricks, implementing bronze, silver, and gold data layers to ensure data accuracy, consistency, and performance in large-scale analytics.
- ✓ Implemented Delta Lake architecture on **ADLS Gen2**, leveraging Hierarchical Namespace and RBAC for secure and efficient data management.
- ✓ Implemented **Delta Live Tables** for seamless continuous data processing and leveraged **Databricks SQL** to perform interactive querying, enhancing data availability and facilitating real-time analytics for informed decision-making.
- ✓ Implemented **Azure Data Lake Storage Gen2** Lifecycle Management policies and Immutable Storage for **Data governance**, resulting in a scalable and cost-effective analytics platform.
- ✓ Integrated **Functional Apps** within Azure Data Factory pipelines for automated data processing tasks, enhancing overall system flexibility and responsiveness.
- ✓ Implemented data integration and transformations within **Azure SQL Server**, ensuring high availability and performance for enterprise-level applications.
- ✓ Designed and implemented scalable data governance frameworks using **Databricks Unity Catalog** to manage permissions, audit data usage, and ensure compliance with industry standards across multiple environments, improving data security and operational efficiency by 30%.
- ✓ Developed Source-to-Target mappings, unit test cases, and automated data validation workflows.
- ✓ Designed real-time data integration solution using **Data Factory** and **Python**, streaming from IoT devices to Data Lake through Databricks, implementing **PySpark** transformations for analytics processing.
- ✓ Integrated **Cosmos DB** with Azure Data Factory and Databricks to create ETL pipelines, facilitating seamless data flow between various sources and Cosmos DB.
- ✓ Conducted a Proof of Concept (POC) on **Microsoft Fabric**, demonstrating its capabilities in streamlining data integration and analytics workflows, resulting in improved data processing efficiency for client applications.
- ✓ Designed and implemented a unified data architecture using **Microsoft Fabric's OneLake** and **Lakehouse** features, enabling data ingestion, storage, and analytics across diverse data sources, for analytical workloads.
- ✓ Implemented **Kafka Connectors** for data ingestion from **POS Terminals** into **Azure Data Lake Storage (ADLS)**, and integrated **Kafka** with **Azure Databricks** for processing of real time data, enhancing ETL workflows and optimizing customer experience analytics.
- ✓ Designed and developed high-availability **MS SQL databases** while optimizing complex **T-SQL queries** for data manipulation and reporting in **Azure SQL Server**.
- ✓ Designed modular, reusable **DBT models** to transform raw data into analytics-ready datasets within **Azure Synapse**, improving data consistency and optimizing processing time for efficient data modelling and analytical workflows.
- ✓ Implemented **Azure Monitor** for comprehensive monitoring of data pipelines, enabling proactive identification and resolution of performance bottlenecks.
- ✓ Implemented **Azure Key Vault** to securely manage secrets, keys, and certificates within Azure Data Factory pipelines, significantly strengthening security and ensuring compliance with industry standards.
- ✓ Collaborated with data scientists to provide processed, high-quality datasets from **Synapse** and **Databricks** for machine learning models, supporting predictive analytics use cases.
- ✓ Created efficient schema designs in **MongoDB**, optimizing collections with indexes for improved query performance in document-based data storage.
- ✓ Experienced in configuring **Linked Services**, creating **Datasets**, and setting up **Integration Runtime** within Azure Data Factory for seamless data integration across diverse environments.
- ✓ Implemented advanced scheduling and automation using Triggers like **Scheduled Trigger**, **Tumbling Window Trigger** and **Event Based Trigger** based on necessity in Azure Data Factory, ensuring timely execution of ETL processes and data synchronization.

- ✓ Developed and maintained real-time data streaming pipelines using **Apache Kafka**, enabling efficient data flow and processing for real-time analytics.
- ✓ Optimized data pipelines and **Spark jobs** in **Azure Databricks** through advanced techniques like Spark configuration tuning, data caching, and data partitioning, resulting in superior performance and efficiency.
- ✓ Developed automated data quality checks and executed large-scale data processing tasks using **PySpark** in **Azure Databricks**, significantly enhancing data trustworthiness and improving processing performance by reducing execution times.
- ✓ Developed and optimized **SQL queries** for data extraction, transformation, and loading (ETL) processes across various platforms, ensuring data accuracy and efficiency.
- ✓ Leveraged **SQL** and **Python** for querying and managing data within data warehouses, enhancing data availability and supporting analytical requirements.
- ✓ Automated infrastructure provisioning on Azure using **Terraform**, managing resources like Data Factories, Storage Accounts, and Databricks clusters, while developing reusable modules that standardized deployments and reduced provisioning time by 40%.
- ✓ Designed and implemented CI/CD pipelines using **Azure DevOps** and **GitHub Actions** to automate the deployment of data pipelines and infrastructure, ensuring efficient and reliable operations.
- ✓ Streamlined infrastructure provisioning with **Azure Repos** and automated end-to-end testing for data pipelines, enhancing version control, collaboration, and deployment quality.
- ✓ Managed and optimized data storage and processing using file formats like **Parquet, ORC, JSON, Avro, and CSV**, ensuring compatibility and efficiency across systems and improving data retrieval speeds.
- ✓ Automated Azure resource management using **PowerShell scripts**, reducing manual configuration time by 40% and improving deployment efficiency.
- ✓ Developed **Star Schema** data models for OLAP systems, ensuring efficient query performance and simplified data navigation.
- ✓ Developed interactive **Power BI dashboards** to visualize key business metrics, improving decision-making processes and enabling real-time insights for stakeholders.
- ✓ Integrated **Power BI** with **Azure Synapse Analytics** for real-time data visualization, reducing the time to insight by 40%.
- ✓ Worked in a fast-paced **Agile environment**, collaborating with cross-functional teams to deliver high-quality data solutions **on time and within budget**.
- ✓ Collaborated with cross-functional teams to develop shared **Confluence** documentation, ensuring alignment and clarity across various data projects.
- ✓ Integrated **GitHub** with **CI/CD** tools to streamline the deployment of data pipelines and applications, ensuring quick and reliable updates to production environments.

Environment: Azure Data Factory, Azure Databricks, Azure Data Lake Storage Gen2, Azure Blob, Snowflake Data Warehouse, Azure Event Hubs, Azure Functions, Azure Logic Apps, Function Apps, Azure Monitor, Cosmos DB, Mongo DB, Azure Synapse Analytics, Azure DevOps, Terraform, Azure Fabric, Azure Key Vault, MS SQL, Oracle, Python, PySpark, JIRA, Kafka, Power Bi.

Client: Arrixon Technologies, Hyderabad, India

Oct 2017 to Dec 2022

Role: Senior Data Engineer

Responsibilities:

- ✓ Designed and implemented complex data integration pipelines in **Azure Data Factory**, utilizing various activities including Copy Activity, Mapping Data Flow, and Control Flow Activities.
- ✓ Configured **Linked Services** to connect to multiple data sources and implemented Triggers for automated execution.
- ✓ Implemented **Azure Data Factory's** Integration Runtime to enable secure data movement between on-premises and cloud environments. Implemented **Self-Hosted Integration Runtime** for accessing data behind corporate firewalls.
- ✓ Optimized Data storage and management in **Azure Data Lake Storage Gen2**, implementing Hierarchical Namespace for efficient data organization.

- ✓ Configured Azure Databricks Workspace to create and manage clusters, developing notebooks for data processing and analysis while implementing ETL processes that leveraged Spark DataFrames and RDDs for efficient data transformation.
- ✓ Implemented **Azure Data Lake Storage Gen2** with **Azure Active Directory (AAD)** integration, ensuring seamless and secure authentication and authorization for robust data access control.
- ✓ Utilized **Python** libraries such as **Pandas** and **NumPy** for data manipulation, transformation, and analysis, enabling efficient handling of large datasets in **Azure Data Lake Storage**.
- ✓ Implemented data validation and cleansing processes in **Python**, ensuring high data quality before loading into **Azure SQL Database** and **Azure Synapse Analytics**.
- ✓ Leveraged **Azure Synapse Link** to enable near real-time analytics on operational data, integrating with **Azure Cosmos DB** for seamless data synchronization.
- ✓ Developed real-time data ingestion solutions using **Azure Data Factory** integrated with **Azure Event Hubs**, improving data synchronization accuracy and reducing latency.
- ✓ Developed real-time data streaming architectures leveraging **Azure Event Hubs**, Databricks, and Synapse Analytics, ensuring instant data accessibility and analysis for enhanced business intelligence.
- ✓ developed **PySpark** applications in **Databricks** to transform 200GB daily data, optimizing ETL workflows from Data Lake to Synapse Analytics, reducing processing costs by 40% through efficient Python.
- ✓ Engineered **Azure Data Factory** pipelines integrating **Data Lake with Databricks**, implementing SQL-based validation checks and Python scripts for data quality monitoring, serving 100+ business users daily.
- ✓ Architected enterprise data platform using Synapse Analytics and Databricks, processing Data Lake sources through PySpark transformations, achieving 99.9% SLA and 45% faster performance.
- ✓ Developed production-grade **Python and PySpark** modules in Databricks for complex ETL operations, orchestrating Data Factory pipelines between Data Lake and SQL databases with 40% improved efficiency.
- ✓ Built automated testing framework using Python in **Azure Data Factory**, validating data quality between Data Lake and Synapse, while implementing PySpark optimizations for 50% faster processing.
- ✓ Implemented delta lake architecture using **Databricks** and PySpark, optimizing Data Lake storage patterns and SQL-based merge operations in Synapse, reducing data processing latency by 60%.
- ✓ Designed and implemented end-to-end ETL pipelines using **Azure Data Factory** and **Python**, incorporating data extraction, transformation, and loading processes that improved data flow efficiency.
- ✓ Spearheaded migration from legacy ETL to **Azure Data Factory**, leveraging **Databricks** notebooks with **PySpark** and **SQL** for Data Lake transformations, resulting in 75% reduction in development time.
- ✓ Migrated and managed large-scale datasets in **Azure Data Lake, and Apache Cassandra**, optimizing storage strategies for performance and cost-efficiency.
- ✓ Designed and developed data models in **Snowflake**, leveraging its MPP architecture for scalable and high-performance data storage and querying.
- ✓ Implemented Data Warehouse solutions leveraging dimensional modeling with **Star Schema** and **Snowflake Schema** for optimized query performance and analytical workflows.
- ✓ Implemented Snowflake's **Time Travel and Zero-Copy Cloning** features for efficient data backup and recovery, reducing downtime and data loss risks.
- ✓ Developed and optimized complex **Snow SQL queries** for efficient data extraction, transformation, and loading (ETL) operations within **Snowflake**.

- ✓ Designed and implemented **Snowflake Schema** for optimized data warehousing, enabling efficient data storage and retrieval.
- ✓ Implemented **Azure Key Vault** integration to manage secrets, keys, and certificates within data pipelines; strengthened security protocols and ensured compliance with organizational policies while reducing potential vulnerabilities by 35%.
- ✓ Managed and optimized data storage and processing using various file formats Like **ORC, PARQUET, AVRO, JSON, CSV**, ensuring compatibility and efficiency across multiple systems.
- ✓ Enabled seamless **Data Visualization and reporting** through integration with **Power BI**, driving data-driven decision-making across the organization.
- ✓ Conducted code reviews using **GIT** to maintain code quality and adhere to best practices, fostering a culture of continuous improvement.
- ✓ Managed project tasks and user stories in Jira, facilitating effective tracking of progress and collaboration within the data engineering team.
- ✓ Utilized SharePoint for document management and collaboration among data engineering team members, ensuring easy access to project documentation and resources.
- ✓ Participated in Scrum ceremonies, including daily stand-ups, sprint reviews, and planning meetings, to foster collaboration and keep the data engineering team aligned.

Environment: Azure Data Factory, Azure Databricks, Azure Data Lake Storage Gen 2, Azure Blob, Logic Apps, Functional App, Snowflake, MS SQL, Oracle, HDFS, Hive, Sqoop, PIG, PySpark, Oozie, Zookeeper, SQL, Python 3.8.2, Scala, Kafka, Power Bi.